ECON 4660 / 8666 — Problem Set 2

International Economic Development · Fall 2026 · Prof. Zhigang Feng
Due Monday, October 5, before class (Canvas). At most 3 pages. No raw data files.

Due *before* the midterm (October 12). The numerical Solow problem and the short definitions are the kind of reasoning the essay will expect. Do not recycle answers from Problem Set 1.

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1. A Solow economy

France lives in a Solow world. The intensive production function is $y = k^{1/2}$. Also $s = 0.2$, $\delta = 0.1$, $n = 0$.

(a) Write the aggregate marginal products of labor and of capital (in terms of K and L).

(b) Capital per worker in steady state.

(c) Output and consumption per worker in steady state.

(d) France had $k = 12{,}000$ in 1965. Using

$$k' = k + sk^{1/2} - (n + \delta)k,$$

find k in 1966.

(e) Growth rate of GDP per worker between 1965 and 1966. Is the economy above or below steady state? Does that match what Solow says about catch-up?

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2. Inequality in a tiny economy

Five people have incomes $2, 4, 6, 8, 20$.

(a) Compute the Gini coefficient (show the formula you use).

(b) In one or two sentences: how is the Gini related to the Lorenz curve? If the richest person transferred $4 to the poorest, would the Gini rise or fall?

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3. Concepts (a few sentences each)

Explain each pair or term. A formula or a labeled sketch is welcome where it helps.

(a) Unconditional convergence vs. conditional convergence.

(b) Poverty vs. inequality.

(c) Demographic transition.

(d) Private vs. social return to education.

(e) [Diagram] Effect of higher household income on desired fertility, and effect of closing the gender wage gap on desired fertility. Label axes.

ECON 8666 only. One extra paragraph: in the education / endogenous-growth discussion, long-run growth depends on parameters such as b and u. Give one concrete policy that could move each parameter, and say in which direction.